

How We Tested

We evaluated the cameras on the following parameters:

Features

We awarded points based on the various features and capabilities of the camera, such as CCD (Charged Couple Device) resolution, maximum image resolution supported, optical and digital zoom, focusing types, metering modes, white balance implementation, flash modes, shutter speed, storage, battery type and video-out capabilities.

Ergonomics

We noted various aspects of the camera's build, and looked for features that enhanced its user-friendliness—aspects such as how comfortable it was to hold, the ease with which the menu can be navigated, the size of the buttons, the placement of the flash and whether it avoids the finger-over-flash syndrome, and the inclusion of special menu-navigation buttons. The size and weight of a digital camera were also taken into consideration.

Performance

Image test: We created a target scene to test the camera's CCD ability to reproduce colour and capture details correctly. The lighting was kept constant. We shot this scene by mounting the camera on a tripod, and chose the highest possible resolution setting. We used the camera's flash and set other settings, such as the white balance, metering and flash modes to Auto mode.

The target scene consisted of a table with flowers that provided a wide tonal range. Gradations in shading and specular highlights were brought out through shiny metallic objects, such as the table clock and the metallic stand with the candle wrapped in plastic. The table also had objects such as curtain lace, a table-cloth, stack of books and the flower vase—all of which provided plenty of minute details. The cameras were graded on the level of accuracy with which each of these details was captured.

For the reference value, we shot the test scene with a professional film camera, and then scanned it on a very high-end scanner, the same one we use to scan all the images published in Digit. Next, we logged the values for each of the three components using the histogram function in Photoshop. The components had the following reference values: red = 147, green = 113, blue = 118

and luminosity = 124. These were compared with the values obtained from each camera to measure the deviation of the images from the optimal reference point.

Details test: To check for the accuracy with which details were reproduced, we divided the entire test scene into three basic zones that helped to analyse three characteristics of the camera. The first zone, comprising varying shades of colour, determined the camera's colour capturing and reproduction capability. The second zone comprised elements with details, such as wrinkles on the face of the doll, and the keyboard text. This allowed us to analyse the camera's focusing ability. The third zone consisted of objects with finer details such as the design of the curtain lace, etc. The exact reproduction of these details gave an indication of the camera's detail capturing ability.

RGB test: To measure colour reproduction, we took a close-up of an RGB circle, printed it on glossy paper, with the flash switched off. The test RGB circle image was created in QuarkXpress so as to get true red, green and blue colours, wherein the Histogram function in Photoshop shows the value as 255 for each colour.

Zoom test: To test the zooming capability of the digicams, we employed the 3X and 6X zoom cameras to take photographs of a stationary target placed at approximately 15 feet. Based on these photographs, we then analysed how close the cameras could zoom onto the subject to provide clean, non-granulated images.

Actual person test: In the actual person test, we shot the photograph of a person under uniform lighting conditions with the camera mode set to Auto. This yielded the real life performance of a camera under indoor lighting. In the final image, we looked for proper skin tone reproduction, white balance, uniform tonal reproduction under florescent lighting, etc.

Value for money

This is evaluated based on the performance and features offered by the camera as compared to its price. Therefore, greater the performance and features, and lower the price, higher will be the value for money offered by a camera. We computed this value by dividing the sum of the camera's performance and features by its price to obtain a value for money index.



- 1 Flowers, and the small ornament with variety of colours on it, was used in the test scene to test the colour reproduction of the digital cameras.
- 2 The Digit CDs, the keyboard and the wrinkles of the face of the doll tested the cameras ability to focus across the length of the frame uniformly.
- 3 For testing the cameras ability to pickup details, we introduced the metallic figure with shining thread around its neck and waist and the lace curtain as the background.
- 4 The metallic table pen stand, with clock and the glass filled with green liquid were used to check the camera capability in reproducing the specular effects (i.e the shiny effect created by the reflection of light on metallic and glass objects).
- 5 Here, we checked the camera's ability to pick up the variation from dark to light regions.